

1. Data split & validation

Train: Jan-Oct 2025 (48,000 rows). Final test: Nov-Dec 2025 (12,000 rows, labels hidden). Because market_index drifts (train mean 1.08 vs validation 0.93) and rates trend by month, a random split would leak the future and give optimistic scores. We use a temporal expanding window with 3 folds:
Fold1: train Jan-Apr -> test May-Jun
Fold2: train Jan-Jun -> test Jul-Aug
Fold3: train Jan-Aug -> test Sep-Oct (closest proxy for Nov-Dec).
Metrics per fold: MAE / RMSE / MAPE / MedAPE. Model selection on mean MAE/MAPE.

2. Data-quality issues & fixes

- weight: 300 NaN (flag + median), 292 negative -> abs() (sign-entry error), 1204 capped at 47500 (flag).
- market_index: 374 NaN train / 249 val -> same-day median from the same table's own calendar (the Jan-Oct train map contains no Nov-Dec date, so validation/test folds use their own days; market is an input feature, not the label, so this is not leakage), fallback global median (+ missing flag). Within-day noise ~0.025 so daily median denoises safely.
- distance: 48 rows floored at 70.0 (flag).
- posted_rate: ~1.4% outliers (x3 or /3) -> kept; robust L1 loss on log-target handles them.
- coordinates are not real geography (e.g. Phoenix shifted); used as lane embeddings, not maps.
- 8 unseen validation cities (Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego): no city-ID features, coordinates only.

3. Key findings

- distance <-> rate corr 0.91: the dominant driver.
- market_index has a huge weekly cycle (Thu ~1.18, Sun ~0.99) but moves prices only ~+-1%%; raw index is a trap, daily-median smoothing (mi_smooth) is used.
- quote_signal is U-shaped vs \$/mile (edges 3.2-3.4, middle 2.1): strong non-linear signal, qs_dev=|q-median| added.
- monthly \$/mile drifts 2.10 (Jan) -> 2.33 (Jun) -> 2.24 (Oct): independent time trend + trend/dow features.

4. Model choice

LightGBM regression_l1 on log1p(posted_rate). GBM captures the U-shaped quote effect and lane interactions that linear models miss; log-target stabilizes skew; L1 ignores label outliers (RMSE ~630 for all models is outlier-driven). Coordinates + 3-way equipment one-hot generalize to new cities.

5. CV results (temporal folds)

fold	model	MAE	RMSE	MAPE	MedAPE
1	naive	206.18	654.18	11.01	7.05
1	ridge-log	104.26	631.50	4.30	1.74
1	hgb-logL2	134.50	640.68	5.75	2.79
1	lgbm-logL1	119.32	636.41	4.83	1.82
2	naive	222.73	662.87	10.11	6.70
2	ridge-log	140.93	624.40	6.01	3.41
2	hgb-logL2	146.14	628.19	6.25	3.19
2	lgbm-logL1	121.91	624.71	5.31	2.19
3	naive	229.10	670.66	10.49	6.80
3	ridge-log	110.70	634.69	4.78	1.98
3	hgb-logL2	119.76	637.91	5.18	2.32
3	lgbm-logL1	106.61	633.91	4.67	1.91

Mean over folds:

	MAE	RMSE	MAPE	MedAPE
model				
hgb-logL2	133.47	635.59	5.73	2.77
lgbm-logL1	115.95	631.67	4.94	1.97
naive	219.34	662.57	10.53	6.85
ridge-log	118.63	630.20	5.03	2.38

Winner: lgbm-logL1 (mean MAE 115.95, MAPE 4.94) vs ridge-log (118.63 / 5.03), hgb-logL2 (133.47 / 5.73), naive

Freight Rate Prediction - Spotter ML Assessment

median \$/mile (219.34 / 10.53). Fold3 (Sep-Oct, best Nov-Dec proxy): lgbm MAE 106.6 / MAPE 4.67. Final model retrained on all Jan-Oct with a time-tail eval split + early stopping (seed 0).

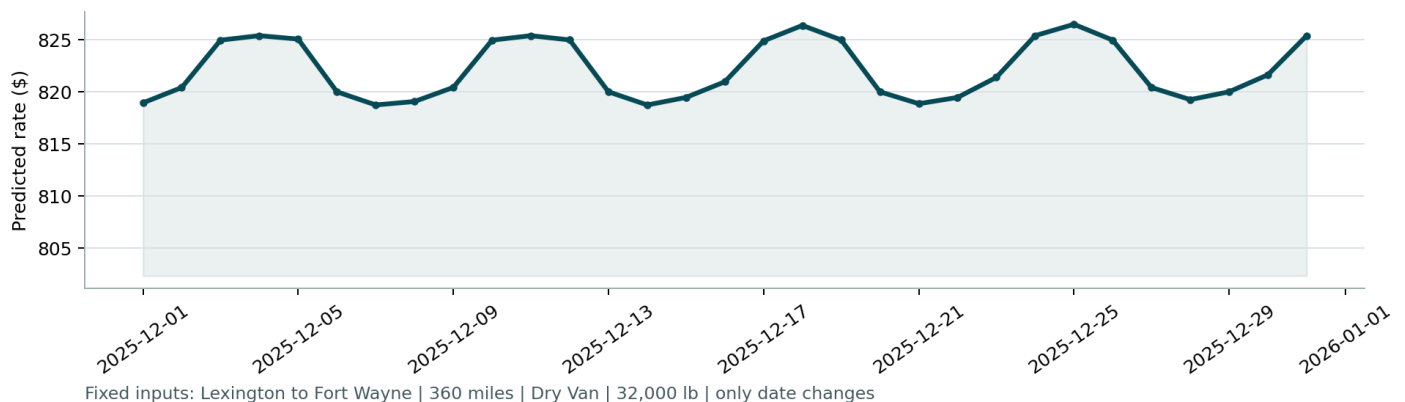
6. December chart (fixed lane)

Lane Lexington -> Fort Wayne, 360 mi, Dry Van, 32,000 lb; only date varies. december_chart_inputs.csv ships WITHOUT market_index / quote_signal / coordinates, so:

- coords: train city medians (both cities exist in train).
- market_index + mi_smooth: calendar-day means from validation.csv (input features covering all Dec dates, not labels).
- quote_signal: December median from validation (2.0512).

Prediction range ~\$819-826 with a small weekly cycle on top of the trend level. Limitation: 10 months cannot separate trend from seasonality, and holiday December is out-of-distribution; chart shape is more reliable than absolute level.

Candidate: December 2025 Predicted Load Rate



7. Post-baseline experiments (not shipped)

After freezing the submission we tested CatBoost on the identical 3 folds and features (catboost 1.2.10, seed 0, time-tail eval split + early stopping):

CatBoost log-target + MAE loss: mean MAE 126.2 (folds 134.6 / 124.5 / 119.6).

CatBoost log-target + RMSE loss: mean MAE 114.8 (folds 128.8 / 103.4 / 112.3).

Lesson: the log transform already converts absolute errors into relative ones ($\log y - \log \hat{y} = \log(y/\hat{y})$) and damps the corrupted labels, so L1 adds little while being harder to optimize - L1 has a constant gradient (± 1) and zero Hessian, which starves the Newton-style leaf splits that symmetric (oblivious) trees rely on, hence ~1000 slow iterations vs ~300 for RMSE. CatBoost-RMSE is marginally best (114.8 vs 116.0, ~1.0%), but on a single seed that gap may be noise, so the validated LGBM-L1 pipeline stays the submission. A 50/50 LGBM + CatBoost-RMSE average is the natural next step: leaf-wise/L1 vs symmetric/L2 gives structurally diverse residuals.

RMSE decomposition (Fold3, submitted model, 9,523 rows): all rows MAE 106.6 / RMSE 633.9 / MAPE 4.67%; after dropping only the worst 1.5% absolute errors: MAE 50.7 / RMSE 74.1 / MAPE 2.33%. The 8 worst rows pair true rates of 13k-20k dollars against sane predictions of 3k-6k (about 2 \$/mile on normal 1,400-3,000 mile hauls): corrupted labels (x3-x5), not model errors. Conclusion: RMSE ~630 is a property of the labels, handled by L1+log robustness during training; deleting high-error rows would be tuning by deletion, so the metric is reported honestly as-is.

8. Reproduce

```
pip install -r requirements.txt
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```
python -m src.validate
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python -m src.predict
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```
python score.py --predictions validation_predictions.csv --december-predictions data/december_chart_inputs.csv
```